Eartn and Planetary Science Letters, 24 (1975) 377–384
© North-Holland Publishing Company, Amsterdam – Printed in The Netherlands

PALEOMAGNETISM AND POTASSIUM–ARGON AGE OF ST. PAUL ISLAND, SOUTHEASTERN INDIAN OCEAN: CONTRASTS IN GEOMAGNETIC SECULAR VARIATION DURING THE BRUNHES EPOCH

N.D. WATKINS

Graduate School of Oceanography, University of Rhode Island, Kingston, R.I. (USA)

I. McDOUGALL

Research School of Earth Sciences, Australian National University, Canberra, A.C.T. (Australia)

and

J. NOUGIER

Department of Geology, Federal University, Yaounde, Cameroon (West Africa)

Received July 31, 1974
Revised version received October 22, 1974

A total of 135 oriented cores have been collected from 18 separate basaltic bodies on St. Paul Island on the axis of the mid-Indian ridge, and subjected to paleomagnetic and K/Ar analyses. Like Amsterdam Island, 70 km to the north, St Paul is restricted to the Brunhes epoch (t = 0–0.69 m.y.) in outcrop. An average volcanic production rate of 10^4 m^3 per year is suggested for that part of St. Paul above sea level. Of the two islands, St. Paul has a much higher angular dispersion of virtual geomagnetic poles (S_F). Because of this, it is concluded that despite their restricted age ranges, St. Paul and Amsterdam are likely to have minimal overlap in age. As support for this interpretation, data from a nearby deep-sea core, which has a Brunhes epoch record spanning 17 m of sediment, is consistent with a highly variable geomagnetic secular variation during the Brunhes epoch.

1. Introduction

St. Paul and Amsterdam (Fig. 1), the only parts of the mid-Indian ridge axis which are above sea level, are amongst the world's most isolated islands, thus providing the only simple means to observe geological properties in a large region.

Amsterdam Island has now been relatively well studied, as the result of mapping and collections made during brief visits in 1968 and 1969, and as the result of the first geological expedition to the island in 1972. Gunn et al. [1] have described the island's geomorphology and the geochemistry which suggests an early stage of evolution from a tholeiitic to an akaline volcano. The paleomagnetic data have been published [2] providing evidence for at least two excursions of the magnetic field during the Brunhes epoch (t = 0–0.69 m.y.), and values for the angular dispersion of virtual geomagnetic poles (VGP) consistent with the present average behaviour of the geomagnetic field in the same latitudes.

Following the expedition to Amsterdam Island in 1972, a five-day visit was made to St. Paul Island (Fig. 2), which is uninhabited. During this period, a series of studies were made, including the sampling of 135 oriented cores from the total of 18 accessible bodies (Fig. 2). The geomorphology, petrography, and geochemistry have been described by Gunn et al. [3], who also summarize the results of earlier geological visits to the island by Zirkel [4], Velain [5], and Phillipi [6]. Briefly, the island is a volcanic cone with a perfectly circular central caldera, breached to the sea by a large normal fault, which has also created coastal cliffs (Fig. 2). The axis of the mid-oceanic ridge, with

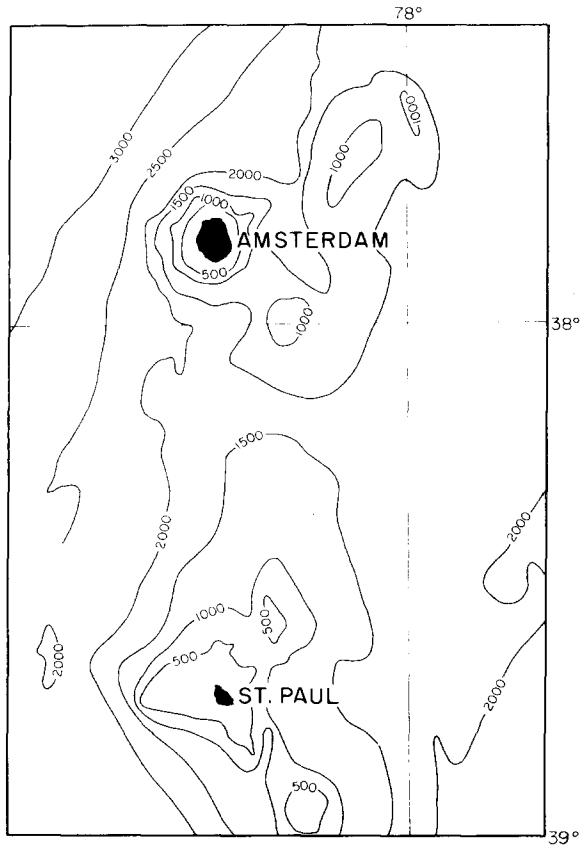


Fig. 1. Bathymetric map of the Amsterdam–St. Paul Platform. From Berthios et al. [25]. Contours in meters below sea level. Latitude in degrees south, longitude in degrees east.

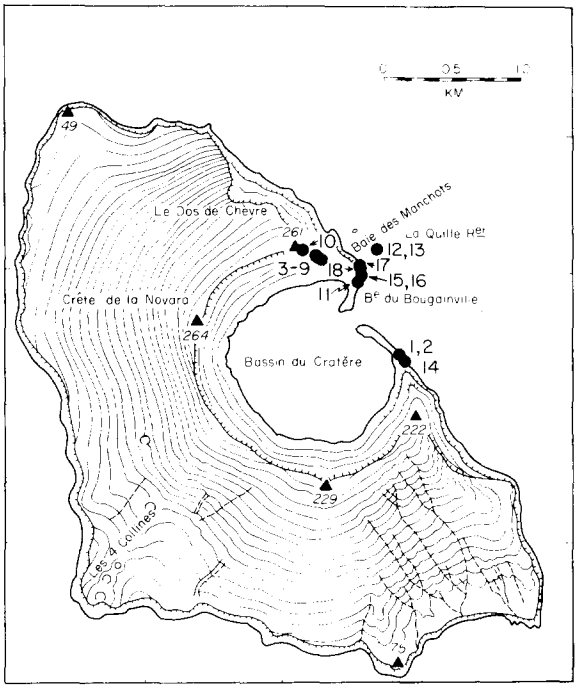


Fig. 2. Topographic map of St. Paul Island, showing samples sites (1 to 18). Contour interval 10 m except on crater wall and cliffs. Solid triangles represent spot heights (meters) above sea level.

characteristic magnetic and bathmetric expression, passes only 3 km to the northwest of the island [7]. The core of the island, exposed at the base of the northeast cliffs, is a series of explosive pyroclastics of "rhyolitic" composition [5], grading up into a series of lava flows draping over the basement. Girod et al. [8] have demonstrated that the lavas have tholeiitic affinities, and Gunn et al. [3] have evidence that the lava flows feature a feldspar fractionation series. Hedge et al. [9] have incorporated some of the Amsterdam and St. Paul samples into a comparative study of strontium isotope ratios of volcanic rocks in the Indian Ocean, and find $^{87}Sr/^{86}Sr$ values ranging from 0.735 to 0.7040.

In this paper we present the paleomagnetic properties and some K/Ar age determinations for the St. Paul collection. The paleomagnetic data will be compared with those from Amsterdam Island.

2. Methods

Field. Up to eight separate cores of 2.5 cm diameter and average length 12 cm were drilled from each of 15 separate lava and 3 minor intrusive bodies on the island, using a gasoline-powered portable drill. Spacing between cores in each body averaged 5 m. Samples were oriented in geographic coordinates to a precision of ±2.5° while still attached to outcrop. The locations of the sample sites are given in Fig. 2.

Laboratory. Directions of remanent magnetism before and after treatment in peak alternating magnetic fields of 100 and 200 oersteds were measured using a spinner magnetometer.

Samples were selected for K/Ar dating using two criteria: the bodies involved should be in known stratigraphic sequence, and the samples should be fresh and unaltered. The method of K/Ar age determinations have been described in detail elsewhere [10]. Potassium is determined by flame photometry, and argon is measured by isotope dilution on samples of 15–35 g.

TABLE 1

Paleomagnetic data for each site

No.	N	D	I	R	K	α_{95}	ϕ'	θ'
1	8	336.8	−62.9	7.914	81.6	6.17	322.1	71.9
2	8	353.5	−66.7	7.989	624.1	2.22	279.1	78.6
3	8	12.8	−49.1	7.981	374.1	2.87	131.4	76.3
4	8	10.4	−53.5	7.946	130.7	4.86	141.2	80.4
5	8	44.8	−60.4	7.979	340.6	3.01	186.3	56.0
6	8	7.6	−76.5	7.902	71.7	6.59	249.8	64.1
7	8	342.7	−62.4	7.991	770.3	2.00	321.2	76.2
8	8	314.6	−66.7	7.976	289.0	3.26	314.2	56.3
9	8	21.6	9.9	7.949	137.8	4.74	106.7	41.9
10	8	343.1	−56.9	7.469	13.2	15.08	347.4	76.7
11	8	333.7	−58.0	7.964	196.4	3.96	339.4	69.6
12	8	324.1	−27.3	7.969	226.7	3.69	14.9	50.1
13	8	333.5	−33.5	7.933	103.9	5.46	21.4	59.2
14	8	9.2	−53.5	7.959	171.4	4.24	137.5	81.2
15	7	336.1	−32.5	6.988	493.8	2.72	26.0	60.3
16	8	350.7	−55.9	7.935	108.1	5.35	2.2	82.3
17	8	322.3	−52.8	7.944	124.2	4.99	345.6	59.3
18	8	356.0	−35.2	7.424	12.2	16.50	66.1	70.3

Explanation of symbols: No. = site number (see Fig. 2). N = total number of separately oriented specimens for given lava. D and I = mean declination and inclination of remanent magnetism in degrees east of north, and below horizontal, respectively (obtained from unit vector per body); R = resultant vector (unit vector per site); $K = (N-1)/(N-R)$; α_{95} = semi-vertical angle of 95% confidence core; ϕ' and θ' = longitude and latitude of mean virtual geomagnetic pole, obtained using unit vector per specimen. All paleomagnetic data are for specimens treated in alternating demagnetizing fields of 100 or 200 oersteds.

The ^{38}Ar spike is inserted using a gas pipette system. The isotopic composition of the extracted argon is measured using a modified AEI MS10 mass spectrometer. The precision of each age measurement is given as two standard deviations based on statistically derived uncertainties for each potassium and argon measurement [11].

3. Results

Paleomagnetism. Paleomagnetic data for each body are given in Table 1. The results of statistical analyses for comparison with the requirements of conventional geomagnetic field models [12] in these latitudes are given in Table 2. Included in Table 2 are the previously published data for Amsterdam Island, and the results of analyses of the combined data sets.

Petrography. Three lavas in known stratigraphic sequence from the inside of the caldera have been selected for dating. These are, in decreasing stratigraphic age, lavas 3, 4, and 7 (Fig. 2). Lava no. 3 is the most prominent lava on the island. It is a 30 m thick, vesicular, coarse doleritic rock. Abundant coarse (up to 5 mm) plagioclase and olivine occur in a medium-grained basaltic groundmass which contains a few percent of interstitial poorly crystallized mesostasis. Lava no. 4, immediately overlying no. 3 is a 6-m thick, feldsparphyric basalt with some vesicles. About 30% of the rock consists of plagioclase phenocrysts, which are set in an extremely fine-grained basaltic groundmass with several percent of interstitial glass. Lava no. 7, which is three lavas below the rim of the caldera, is a 2-m thick, aphyric, dense, dark basalt. Rare plagioclase microphenocrysts occur in a cryptocrystalline groundmass. The rock is fresh and virtually holocrystalline.

K/Ar ages. Results of the age determinations are given in Table 3.

TABLE 2a

Fisher (1953) statistical analyses of paleomagnetic directions and VGP

Island(s)	B	N	D	I	R	K	α_{95}	ϕ'	θ'
Analysis of directions									
St. Paul	17	135	347.3	−55.1	16.1585	19.0	8.4	1.0	79.4
Amsterdam	27	189	3.2	−58.0	26.3290	38.7	4.5	188.1	87.3
Combined	44	324	356.3	−57.0	42.3257	25.7	4.3	350.3	87.1
Analysis of VGP									
St. Paul	17	135			15.8410	13.8	10.0	349.2	80.6
Amsterdam	27	188			26.09	28.5	4.3	205.5	86.6
Combined	44	323			41.6751	18.5	5.1	307.9	87.2

B = number of separate sites; N = total number of specimens; D and I = mean declination and inclination of remanent magnetism (D in degrees east of north; I in degrees with respect to horizontal, negative being upward); R = resultant vector; $K = (N-1)/(N-R)$; α_{95} = semi-vertical angle of 95% confidence cone; ϕ' and θ' = longitude and latitude of mean virtual geomagnetic pole (in degrees east and north, respectively). All data are for specimens treated in alternating fields of 100 and 200 oersteds, and minimum dispersion computation for each site. All parameters are for unit vector per site, rejecting data when VGP site latitude < 45°.

4. Discussion

4.1. Potassium–argon dating

The proportion of radiogenic argon detected in samples from lavas SP 4 and SP 7 was less than 1%, so that the reported ages must be regarded as only approximate, as indicated by the quoted errors. Nevertheless it can be stated with a considerable degree of confidence that these lavas are less than 0.1 m.y. old, confirming the youthfulness of the volcano.

The much older apparent K/Ar ages, averaging

TABLE 2b

Angular standard deviation VGP

Island(s)	S_T	Kt	Kw	Ka	K_f	S_F	Su	Si
*Computation of S_F with respect to mean VGP**								
St. Paul	22.1	13.4	45.5	2100.0	14.0	21.7	28.2	17.6
Amsterdam	15.3	28.2	34.5	2100.0	32.4	14.2	17.2	12.1
Combined	19.1	18.1	44.4	2100.0	19.3	18.4	21.4	16.2
Computation of S_F with respect to present spin axis								
St. Paul	24.1	11.3	45.5	2100.0	11.7	23.7	30.8	19.2
Amsterdam	15.6	26.8	34.5	2100.0	30.6	14.6	17.7	12.5
Combined	19.2	17.1	44.4	2100.0	18.9	18.6	21.7	16.3

S_T = total dispersion, where $S_T^2 = 1/(B-1)\ \Sigma_{i=1}^{B}(\delta i)^2$, and δi = angular difference between ith site VGP and reference pole; B = number of sites; $Kt = 2(S_T)^{-2}$; Kw = within-lava precision parameter for VGP, where $Kw = \Sigma_{N=1}^{K}(Ni-1)/(Ni-Ri)$ and Ni = number of specimens in ith lava, Ri = resultant vector to ith lava applying unit vector per specimen VGP; Ka = precision parameter due to effect of magnetic anomaly existing during lava emplacement at the site (see Doell [23] and Ellwood et al. [16] for discussion); K_f = precision parameter reflecting ancient geomagnetic field variation, where $1/K_f = 1/Kt - 1/NKw - 1/Ka$, where N = average number of specimens per site; $S_F = (2/K_f)^{1/2}$; Su and Si = upper and lower 95% confidence parameters for S_F [24]. Data for bodies with VGP latitudes > 45° only.

* See Table 2a for mean VGP.

TABLE 3

K/Ar age determination (see Fig. 2 for site locations)

Sample No.	Laboratory No.	K wt.%	Radiogenic ^{40}Ar (10^{-13} mol g^{-1})	100 radiogenic ^{40}Ar / total ^{40}Ar	Calculated age ± 2 s.d. (m.y.)
SP 7	72-494	0.864 0.865	0.614	0.8	0.04 ± 0.03
SP 4	72-493	0.428 0.428	0.163	0.3	0.02 ± 0.01
SP 3	72-492	0.383 0.384	2.852 3.909	2.9 2.7	0.42 ± 0.16 0.57 ± 0.07

Constants used: $\lambda_e = 0.585 \times 10^{-10}$ yr^{-1}; $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}; $^{40}K/K = 1.19 \times 10^{-2}$ atom%.

about 0.5 m.y., for the coarse doleritic lava SP 3 would appear to be inconsistent with the ages determined on the overlying lavas because no obvious erosional break was evident in the lava sequence. It is possible that this rock contains excess argon, occluded within the large plagioclase and olivine crystals, which probably crystallized prior to eruption of the magma as a lava flow. If excess radiogenic argon was present in the rock at the time of its eruption, then a basic assumption made in calculating the K/Ar ages is invalid and would result in model or apparent ages that are greater than the true age. Because of the large difference in apparent age between the doleritic lava and the overlying lavas we prefer such an interpretation, to the alternate explanation: namely, the existence of a substantial hiatus between eruption of lavas SP 3 and 4.

The youth of the island is consistent with its proximity to the ridge axis.

4.2. Volcanic production rates

When the volume of the island above sea level is considered, estimates of volcanic production rates are possible. If an age of 0.1 m.y. is considered as correct for the maximum age of outcrop, then an average volcanic production rate in the order of 10^4 m^3 per year is apparent. This would rise to 10^5 m^3 per year if an age of only 0.10 m.y. is assumed. These figures are three orders of magnitude less than lava production rates for some other islands and plateau basalts [13], so St. Paul Island cannot be considered to be a particularly active volcano.

4.3. Paleomagnetic properties

Within-body precision is, with only two exceptions, high. The bodies sampled are all normal polarity, consistent with the Brunhes epoch age. On St. Paul, there are no lavas with low-latitude VGP positions, which have been detected, and interpreted as evidence for departures of the geomagnetic field, on Amsterdam Island. The mean VGP is, at the 95% confidence level, coincident with the geographic pole. This is evidence against the validy of Wilson's [14] offset dipole hypothesis (which would require a VGP 8° to 9° beyond the geographic pole from the island in the 180° to 270° longitudinal quadrant), but is consistent with the unequal co-axial dipole pair model [15] in which the apparent dipole offset on high southern latitude sites is minimal, compared to that in northern latitudes.

The angular dispersion of VGP (S_F) is, at the 95% confidence level, higher than that for the geomagnetic field model based on the present geomagnetic field (Fig. 3; see caption to Table 2; and [16] for summary of definitions and models). This is surprising, since similar analysis of paleomagnetic data from Amsterdam Island [2], the results of which are included in Table 2, provided a value of S_F consistent with present geomagnetic field behaviour. It would on first analysis appear that the contrast in values of S_F could be explained by one of two major possibilities:

(1) S_F for St. Paul is the best value for the region, and the lower value for Amsterdam is anomalous, being due to an insufficient sampling of the full geo-

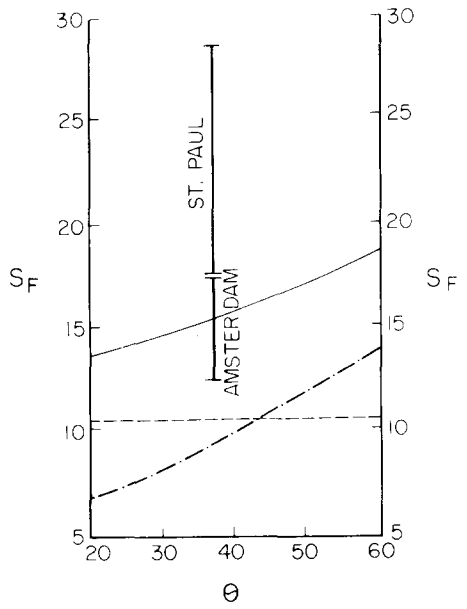


Fig. 3. Diagram showing angular dispersion (S_F) of virtual geomagnetic poles versus latitude (θ) for St. Paul and Amsterdam Island data (Table 2), superimposed on latitude dependence of S_F for geomagnetic field models A (alternating dashes and dots = non-dipole field only); B (dashes = dipole wobble only); and S_T (unbroken line = resultant of A plus B). The data are plotted in the form of the 95% confidence limits [24]. See Doell and Cox [12] for further discussion of models. The diagram shows that at the 95% confidence level, S_F for St. Paul is higher than that of Amsterdam.

magnetic secular variation spectrum, because of rapid extrusion of lavas. This is unlikely, since Amsterdam is geomorphologically much older than St. Paul. Were S_F to be significantly higher than that for the present field, than it could be interpreted to be support for the existence of a substantial standing component in the non-dipole field [17], as is suspected to be a possible explanation for paleomagnetic results from Marion Island [18].

(2) S_F for St. Paul is anomalously high, because of a fortuitous coincidence of lava extrusion during periods when the geomagnetic field was least in conformity with an axial dipole configuration. We do not favour this possible interpretation since there does

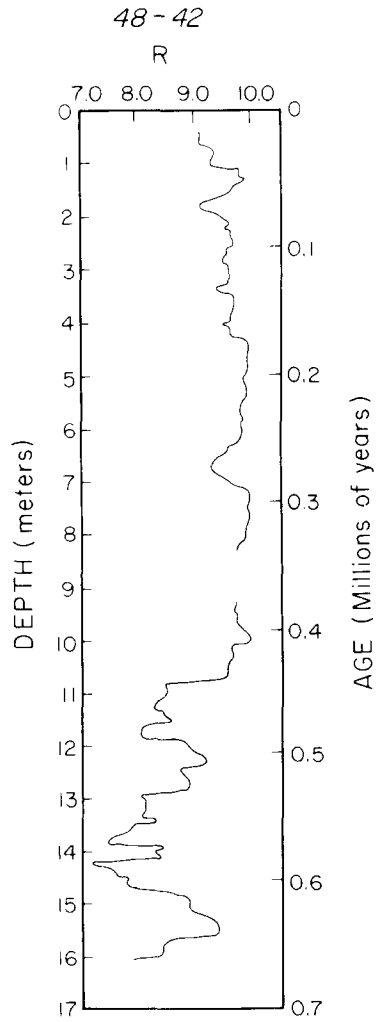


Fig. 4. Variation of the resultant vector for the paleomagnetic inclination of 10 successive specimen on a sliding average basis (applying unit vector specimen) in "Eltanin" core 48-42. This deep-sea sedimentary core was collected at latitude 31°19.0'S; longitude 93°27.5'E, in 2495 fathoms of water. The total length of the core is 24.68 m, the top 17.0 m of which is Brunhes in age. Specimens were taken at 10-cm intervals. The age (right-hand scale) is assigned on the assumption of a constant sedimentation rate. The purpose of the diagram is to demonstrate the possibility that the Brunhes epoch geomagnetic field directions were much more variable (smaller resultant vector) between t = 0.6 and 0.4 m.y., and least variable between t = 0.4 and 0.2 m.y. (large resultant vector). Similarly, a period of greater variability may have occurred since t = 0.04 m.y. If this interpretation is correct, then the contrast in S_F for Amsterdam and St. Paul (Fig. 3) could be due to age differences within the Brunhes, but it is stressed that bottom dynamic processes could be involved in creating variation in R [22]. Note that the variation of paleomagnetic declination, which could also effect R, could not be considered, because of the lack of azimuthal orientation of the core.

not exist any grouping of VGP, which are in fact distributed through all longitudinal segments (Table 1). Although almost all the bodies sampled on St. Paul appear to be restricted to the last 0.1 m.y., this does not appear to be a factor in causing the high value of S_F, since restricted time ranges would most reasonably be expected to result in anomalously low values of S_F.

There exists no simple explanation for the contrasting values of S_F for the respective islands. We are forced to propose, therefore, that the volcanic activity in St. Paul and Amsterdam, although both restricted to the Brunhes epoch, does not overlap in time; and that the two periods involved were of contrasting geomagnetic secular variation, similar to, and greater than, present secular variation values, respectively.

In order to investigate whether periods of highly variable geomagnetic secular variation (and thus variable S_F values) may have existed during the Brunhes epoch in this region, paleomagnetic data from "Eltanin" core 48-42 have been examined. This deep-sea sedimentary core, which contains the largest apparently continuous Brunhes epoch record obtained during the entire "Eltanin" piston coring program [19,20], was taken from the Diamantina Trough, 1500 km northeast of St. Paul and Amsterdam, and should therefore have recorded a similar geomagnetic secular variation to that which the islands experienced. Although the recording of geomagnetic behaviour in deep-sea sediments is obviously subject to imperfections [21,22], within-core comparisons of paleomagnetic data should provide the best possible measure of variations of S_F with time, if subjected to filtering to minimize very short period changes, or other noise. In Fig. 4, the variation of the resultant vector (R) of the inclination of remanent magnetism for ten successive specimens is shown. One interpretation of these data is that between t = 0.6 and 0.4 m.y., the direction of the geomagnetic field was much more rapidly varying than at other times represented. Similarly, shorter periods of increased variation occurred since t = 0.04 m.y., whereas the field was relatively unvarying between t = 0.4 and 0.2 m.y. While it is fully realized that other factors could explain the results [22], these data are nevertheless capable of supporting an interpretation for the contrasting St. Paul and Amsterdam S_F values as a reflection of formation between t = 0.04 and 0.0 m.y., and t = 0.4 and 0.2 m.y., respectively.

Acknowledgements

This work was made possible by the generosity of M. Pierre Rolland (administrator of the Terres Australes et Antarctiques Francaises) and M. Marcel Bloch (chief scientist of TAAF) who provided transport to St. Paul, and all logistic support on the island. This work is supported by grant GV 32480 form the Office of Polar Programs, National Science Foundation.

References

1 B.M. Gunn, C.E. Abranson, J. Nougier, N.D. Watkins and A. Hajash, Amsterdam Island, an isolated volcano in the southern Indian Ocean, Contrib. Min. Petrol. 32 (1971) 79.
2 N.D. Watkins and J. Nougier, Excursions and secular variation of the Brunhes epoch geomagnetic field in the Indian Ocean region, J. Geophys. Res. 78 (1973) 6060.
3 B.M. Gunn, N.D. Watkins, W.E. Trzcienski, Jr. and J. Nougier, The Amsterdam–St. Paul volcanic province and the formation of low Al tholeiitic andesites, Lithos (1975) in press.
4 F. Zirkel, Rhyolithperlit von St. Paul, Z. Deutsch. Geol. Ges. (1867) 773.
5 C. Velain, Description géologique de la presqu'île d'Aden, de l'île de la Réunion, des isles St. Paul et Amsterdam, Rev. Sci. 14 (1878) 52.
6 E. Phillipi, Geologie der Inseln St. Paul und Amsterdam, Deutsch. Sudpolar Exped. 2 (1909) 367.
7 R. Schlich and P.H. Patriat, Anomalies magnétiques de la branche et de la dorsale médio-indienne entre les iles Amsterdam et Kerguelen, Acad. Sci. Paris Compt. Rend. 272 (1971) 773.
8 M. Girod, G. Camus and Y. Vialette, Sur la présence de tholéiites à l'île St. Paul (Océan Indien), Contrib. Min. Petrol. 33 (1971) 108.
9 C.E. Hedge, N.D. Watkins, R.A. Hildreth and W.P. Doering, $^{87}Sr/^{86}Sr$ ratios in basalts from islands in the Indian Ocean, Earth Planet. Sci. Lett. 21 (1973) 29.
10 I. McDougall and N.D. Watkins, Age and duration of the Reunion geomagnetic polarity event, Earth Planet. Sci. Lett. 19 (1973) 443.
11 I. McDougall, H.A. Polach and J.J. Stipp, Excess radiogenic argon in young subareal basalts from the Auckland volcanic field, New Zealand, Geochim. Cosmochim. Acta 33 (1969) 1485.
12 R.R. Doell and A. Cox, The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle, in: Nature of the Solid Earth, ed. E.C. Robertson (McGraw-Hill, New York. 1972) 245.

13 N.D. Watkins and A.K. Baksi, Magnetostratigraphy and oroclinal folding of the Columbia River, Steens, and Owyhee Basalts in Oregon, Washington, and Idaho, Am. J. Sci. 274 (1974) 148.
14 R.L. Wilson, Permanent aspects of the Earth's non-dipole field over Upper Tertiary Times, Geophys. J. R. Astron. Soc. 19 (1970) 417.
15 N.D. Watkins, Hemispherical contrasts in support for the offset dipole hypothesis during the Brunhes epoch: the case for an unequal co-axial dipole pair as a possible geomagnetic field source, Geophys. J. R. Astron. Soc. 28 (1972) 193.
16 B.B. Ellwood, N.D. Watkins and C. Amerigian, Brunhes epoch geomagnetic secular variation on Terceira Island, central-north Atlantic, J. Geophys. Res. 78 (1973) 8699.
17 T. Yukutake and H. Tachinaka, Separation of the earth's magnetic field into the drifting and the standing parts, Bull. Earthquake Res. Inst., Univ. Tokyo 47 (1969) 65.
18 C. Amerigian, N.D. Watkins and B.B. Ellwood, Brunhes epoch geomagnetic secular variation on Marion Island: contribution to evidence for long-term regional geomagnetic secular variation, J. Geomag. Geoelectr. (1974) in press.
19 N.D. Watkins, U.S.N.S. "Eltanin" cruise 48 to Mid-Indian and Broken ridges, Antarctic J. United States 6 (1971) 269.
20 N.D. Watkins, Marine geology, Antarctic J. United States 8 (1973) 69.
21 N.D. Watkins, Review of the development of the geomagnetic polarity time scale and discussion of prospects for its finer definition, Geol. Soc. Am. Bull. 83 (1972) 551.
22 C. Amerigian, Sea-floor dynamic processes as the possible cause of correlations between paleoclimatic and paleomagnetic indices in deep-sea sedimentary cores, Earth Planet. Sci. Lett. 21 (1974) 321.
23 R.R. Doell, Paleomagnetic studies of Icelandic lava flows, Geophys. J. R. Astron. Soc. 26 (1972) 459.
24 A. Cox, Confidence limits for the precision parameter K, Geophys. J. R. Astron. Soc. 18 (1969) 545.
25 L. Berthios, R. Schlich and Ph. Patriat, Carte bathymétrique du Sud de l'Océan Indien, Cashier de l'Institut de Physique du Globe (Paris) No. 36 (1969) 26 pp., 3rd. ed.